ICML 2023 | Teaching a Fake-Audio Detector New Data Without Forgetting the Old

Regularized Adaptive Weight Modification (RAWM): a replay-free continual-learning method that cuts catastrophic forgetting to roughly a tenth of naive fine-tuning.

A fake-audio detector that scores near-perfectly on its own benchmark can fall apart the moment it hears speech from a different dataset. Trained on one collection of genuine and synthetic utterances, it learns exactly what that collection's spoofing attacks look like, and its Equal Error Rate (EER, the point where false accepts equal false rejects) stays low. Point it at audio recorded elsewhere, generated by newer synthesis tools, and the same detector's EER can jump by two orders of magnitude.

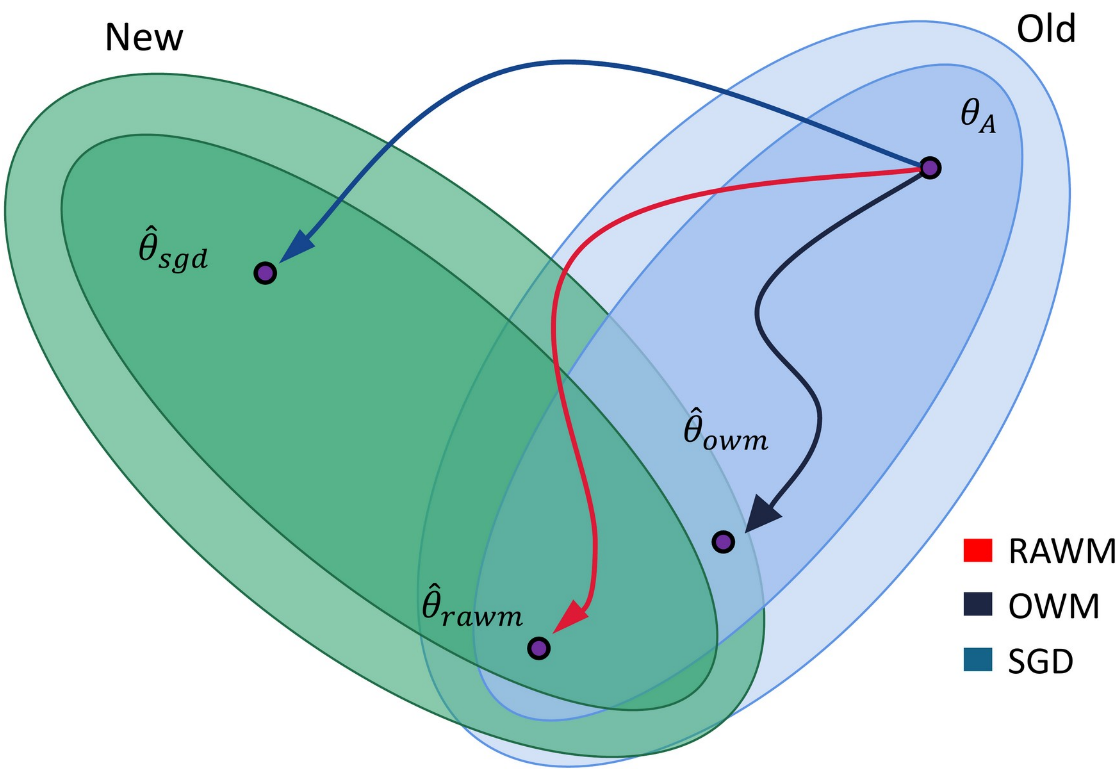

Figure 1. RAWM versus SGD and OWM in the parameter landscape. Plain SGD fine-tuning lands the solution θ_sgd deep inside the New-data region but outside the Old, so old knowledge is lost. OWM's θ_owm reaches the overlap of the two regions but not its best sub-region. RAWM steers the update to θ_rawm, inside the shared sub-region where the model performs well on both old and new datasets at once.

The obvious remedy is to fine-tune on the new data. It works for the new data, and it quietly destroys the old: the network overwrites the features that made it accurate on the original dataset. This is catastrophic forgetting, and it is a real operational problem. Deepfake speech keeps evolving, so deployed detectors must keep absorbing new attack types, yet the original training data is frequently out of reach. A vendor ships a pretrained detector without its private corpus; a public model can be fine-tuned but not re-trained from scratch. You need to add new knowledge without replaying any old samples.

This ICML 2023 paper from the Institute of Automation, Chinese Academy of Sciences and collaborators proposes Regularized Adaptive Weight Modification (RAWM) to do exactly that. The idea has two parts: adapt the direction of each weight update to how genuine-heavy the current batch is, and add a regularization term that keeps the model's old inference behavior intact. No past samples are stored or replayed. Across four fake-audio datasets, RAWM cuts forgetting to about a tenth of naive fine-tuning and halves the error on the new dataset, and the same recipe transfers to speech emotion recognition and image recognition.

Paper: https://arxiv.org/abs/2308.03300

Code: https://github.com/Cecile-hi/Regularized-Adaptive-Weight-Modification

Strong in-domain, brittle across datasets

Fake audio detection has become critical as speech synthesis and voice conversion produce increasingly human-like speech. The catch is generalization. A detector trained on ASVspoof2019LA reaches an EER of 0.258 on its own evaluation set, but its error climbs steeply on audio from other conditions, reaching 91.473 on the In-the-Wild set of real-world deepfakes. Prior cross-dataset fixes such as ensemble learning and domain adaptation lean on samples from the old dataset, exactly what is unavailable once a detector is released without its training corpus.

So the question becomes narrowly practical: how do you fine-tune on new data without forgetting the old, when you cannot keep the old data around? Orthogonal Weight Modification (OWM) already offers a replay-free answer, constraining each update to a direction that does not disturb the subspace spanned by previous inputs. But OWM treats every input the same, and in fake audio detection that leaves value on the table.

The observation RAWM exploits

Genuine speech tends to look similar from one dataset to the next: it is real human audio, and its feature distribution barely shifts across collections. The fake speech is what varies, because each dataset uses different synthesis and conversion attacks. That asymmetry is a usable regularity. If a training batch is mostly genuine, the model is largely looking at content it already understands, so the update should lean toward preserving old knowledge; if a batch is fake-heavy, the update should lean toward learning the new attack. A method that constrains every update identically, as OWM does, cannot express that distinction.

The method: two moving parts

Adaptive weight modification (AWM) is the first part. RAWM starts from OWM's projector P, which points an update away from the subspace of previous inputs. On top of it, RAWM builds a second projector Q that is orthogonal to P and scaled by a factor beta, the ratio of genuine to fake utterances in the batch. The two projectors are normalized and combined into a single modified direction R that the actual weight step follows. Concretely, when a batch is mostly genuine, R tilts toward the knowledge-preserving direction; when it is fake-heavy, R tilts toward learning the new data. The batch composition, not a fixed rule, sets the update direction.

Regularization is the second part, and it handles the harder cases. A few datasets record genuine audio under acoustic conditions so different that the similar-distribution assumption breaks down. For those, RAWM adds a distillation-style term: the frozen pretrained model acts as a teacher, the fine-tuned model as a student, and a modified cross-entropy loss forces the student's softened outputs to match the teacher's. This keeps the old inference distribution alive even when the new dataset sounds acoustically foreign, and the authors find that a coefficient eta = 0.50, weighting old and new data equally, gives the best overall trade-off.

Results: low error on old and new at once

The headline experiment trains a Wav2vec 2.0 based detector through a four-dataset sequence, from ASVspoof2019LA (S) to ASVspoof2015 (T1), VCC2020 (T2), and In-the-Wild (T3), then evaluates on all four. The measure is the good kind of low: EER in percent, lower is better. The pattern in Table 1 is stark. The Fine-tune baseline learns the last dataset but wrecks the source, and even dedicated continual-learning methods leave sizable gaps. RAWM holds the lowest or near-lowest error on every set at once, landing close to the Train-on-All joint-training upper bound that quietly gets to see all data together.

Table 1. Equal Error Rate (EER, %, lower is better) under four-dataset sequential training S -> T1 -> T2 -> T3, evaluated on all four sets. RAWM keeps error low across old and new datasets while the baseline collapses; Train-on-All is a joint-training upper bound.

Method	S	T1	T2	T3
Baseline	0.258	24.532	46.503	91.473
Train-on-All	1.324	0.561	3.579	2.008
Fine-tune	7.068	2.841	5.674	4.543
EWC	5.569	2.444	6.510	5.129
OWM	4.083	2.167	6.480	5.472
LwF	2.714	1.621	4.875	4.325
DFWF	3.476	3.735	7.345	6.114
RAWM (Ours)	1.508	0.641	3.850	3.163

Read the baseline row as the cost of doing nothing and the RAWM row as the cost of RAWM: the source-set error stays near the original 1.508 while the three later datasets settle around 3 to 4, versus a baseline that degrades past 90. Against EWC, OWM, LwF, and the fake-audio-specific DFWF, RAWM is the only method that stays competitive on the source and every target at once rather than trading one set for another one.

It survives the few-sample and cross-task tests

Continual learning is most useful when new data is scarce, so the authors also test a few-sample setting: train on S, then adapt on only 100 samples of T1. Fine-tuning overfits the handful of new examples and forgets S badly, while RAWM learns T1 and barely disturbs the source.

Table 2. Few-sample continual learning: models are first trained on S, then adapted on only 100 samples of T1, and evaluated on both (EER %, lower is better). RAWM recovers T1 while barely disturbing S.

Method	S	T1
Baseline	0.258	24.532
Train-on-All	0.279	0.291
Fine-tune	7.951	0.617
EWC	2.972	0.619
OWM	2.683	0.617
LwF	3.198	0.542
DFWF	1.975	0.733
RAWM (Ours)	0.923	0.312

The mechanism is not specific to spoofing. Because the underlying regularity, some classes staying similar across datasets while others vary, appears in many problems, RAWM also transfers to four-emotion speech emotion recognition trained MSP-Podcast then IEMOCAP, and to the CLEAR-10 image recognition benchmark. On emotion recognition it keeps the highest accuracy among continual-learning methods on both the source and the target.

Table 3. Generalization beyond audio spoofing: four-emotion speech emotion recognition accuracy (%, higher is better), trained MSP-Podcast -> IEMOCAP. Among continual-learning methods RAWM best retains the source while learning the target.

Method	MSP-Podcast	IEMOCAP
Baseline	54.446	30.043
Train-on-All	54.396	57.262
Fine-tune	24.094	50.379
EWC	35.819	48.698
OWM	32.267	48.162
LwF	38.800	44.034
RAWM (Ours)	41.995	54.229

What each part contributes

An ablation separates the two components. When old and new datasets share a similar feature distribution, adaptive weight modification does most of the work, and removing it raises error sharply. When they are recorded under very different acoustic conditions, the regularization term becomes the key to overcoming forgetting. Across the full four-dataset sequence, dropping adaptive weight modification hurts more than dropping regularization, so it is the primary driver and regularization the complement for acoustically foreign conditions.

Why it matters

RAWM's practical appeal is that it fits the constraint real deployments actually face: you can teach a released fake-audio detector new datasets without keeping any of the old data around, and without its accuracy on the original data collapsing. The two cheap ideas behind it, adapting each weight update to how genuine-heavy the batch is and regularizing the model to remember its previous behavior, need no stored samples. Because the same recipe also lifts speech emotion and image recognition, the asymmetry it exploits looks common well beyond fake audio detection.